Proficient in Revit MEP (80 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	**AN INTRODUCTION TO REVIT MEP** Introduction to Revit MEP Design **Graphical User Interface:** Ribbon Tabs Contextual tabs Family Editor tabs Customizing ribbon Quick access toolbar Options Bar Properties Palette Project Browser View Control Bar Status Bar Info center **User Interface Control** Menu and Settings Keyboard Shortcuts Graphics Context Menus	Theory/Demo	2.0Hrs
2	Session 2	Exercises included for Practice	Practicals	2.0Hrs
3	Session 3	**WORKSHARING** **Collaboration** Central files Create a New work set Create a new Local file Synchronizing a Local file with Central file Working with model elements and their worksets Deleting a workset Control visibility and worksets **Work sharing** Working with Linked Revit file Placing Link Revit file Using Shared Coordinates Managing Revit Links Controlling Visibility of Revit Links	Theory/Demo	2.0Hrs
4	Session 4	Exercises included for Practice	Practicals	2.0Hrs
5	Session 5	**Monitoring Elements within Shared Models** Copy/Monitor Copy Elements from a Linked Model Monitor Elements in a Linked Model Copy/Monitor Workflow for Linked Models Reviewing Warnings for Monitored Elements Actions for Coordination Review Stopping Element Monitoring Link CAD Files	Theory/Demo	2.0Hrs
6	Session 6	Exercises included for Practice	Practicals	2.0Hrs
7	Session 7	**HVAC COOLING AND HEATING LOAD ANALYSIS-I** **Space Modeling:** Placing Spaces Space Property Schedule Modify Space Properties Zones Building Construction	Theory/Demo	2.0Hrs
8	Session 8	Exercises included for Practice	Practicals	2.0Hrs
9	Session 9	**Performing Heating and Cooling Load Analysis Supply Airflow schedule:** Load Analysis Weather data Outdoor Air filtration Details Heating and Cooling loads report	Theory/Demo	2.0Hrs
10	Session 10	Exercises included for Practice	Practicals	2.0Hrs
		HVAC COOLING AND HEATING LOAD ANALYSIS-II Logical systems Mechanical settings		

11	Session 11	System Browser **Setting Up Air Systems:** Parameters **Mechanical System & Duct Work:** Air distribution components **Mechanical Equipment Components:** Air Conditioning/Handling Units VAV Boxes **Duct Work:** Create New duct Types Flex Duct Automatic Duct routing Manual Duct routing Duct Justification Duct Sizing Duct add and remove Insulation Add Lining and Remove lining Check systems System Inspector Duct Pressure Loss report Color Fill Legend & Duct Legend	Theory/Demo	2.0Hrs
12	Session 12	Exercises included for Practice	Practicals	2.0Hrs
13	Session 13	**Mechanical Piping System:** Parameters Creating Pipe systems Creating Pipe Types Selecting Fitting for Pipe Types Mechanical pipe settings **Pipe Routing Options:** Automatic & Manual Pipe Routing **Pipe Fittings:** Using Pipe Fitting controls & Placing Fittings System Filters	Theory/Demo	2.0Hrs
14	Session 14	Exercises included for Practice	Practicals	2.0Hrs
15	Session 15	**FAMILY CREATION** Reference Planes and Lines, Constraints and Dimensions **Parameters:** Types of parameters Instance Parameters Parameter Discipline Type, Grouping **3 kind of Family:** System Families Loadable Families In-Place Families Family Editor Tools Type Catalog Family	Theory/Demo	2.0Hrs
16	Session 16	Exercises included for Practice	Practicals	2.0Hrs
17	Session 17	**Solid Modeling:** Extrusions Blends Revolves Sweeps Swept Blends Join geometry Voids Visibility control **Equipment connectors:** Pipe connectors Electrical connectors Duct connectors **Mechanical Equipment Creation**	Theory/Demo	2.0Hrs
18	Session 18	Exercises included for Practice	Practicals	2.0Hrs
		EFFICIENT LIGHTING DESIGN Spaces and Lighting Reflected Ceiling plan		

19	Session 19	Lighting Analysis Light Fixtures Ceiling changes Overhead fixtures Wall – mounted lights Switches **Site Lighting:** Site lighting layout & Analysis	Theory/Demo	2.0Hrs
20	Session 20	Exercises included for Practice	Practicals	2.0Hrs
21	Session 21	**Power and Communications:** Modeling Methods for Power and Systems Devices Using Annotation Symbols Using Face–Hosted Families Avoiding Interference of Symbols Creating Circuits **Creating a Fire Alarm System Model:** Fire Alarm Riser Diagram Fire Alarm Diagram Using Drafting Tools and Symbols **Placing Devices and Equipment Connections:** Disconnect Switches Distribution Equipment and Transformers Switchboards Panels	Theory/Demo	2.0Hrs
22	Session 22	Exercises included for Practice	Practicals	2.0Hrs
23	Session 23	**Creating Power Distribution Systems:** Power Diagrams Modeling Conduit and Cable Tray Defining Electrical Settings Placing Conduit in a Model Placing Cable Tray in a Model Circuiting and Panels Establishing Electrical Settings Wiring Settings Voltage Definitions Distribution Systems Load Calculations	Theory/Demo	2.0Hrs
24	Session 24	Exercises included for Practice	Practicals	2.0Hrs
25	Session 25	**Creating Circuits and Wiring for Devices and Fixtures:** Editing Wiring Editing Circuits Drawing Wires Manually Controlling Wire Display **Managing Circuits and Panels** Circuit Properties Wire Properties Panel Properties	Theory/Demo	2.0Hrs
26	Session 26	Exercises included for Practice	Practicals	2.0Hrs
27	Session 27,28	**Electrical Equipment Creation** **Lighting fixture creation** **Data device with annotation symbol creation**	Theory/Demo	4.0Hrs
28	Session 29,30	Exercises included for Practice	Practicals	4.0Hrs
29	Session 31,32	**PLUMBING** Working with Plumbing Fixtures Copy/Monitor Plumbing Fixtures Sloping Pipe Annotating Invert Elevation and Slope Using Fittings Using Pipe Fitting Controls Placing Valves Setting Visibility of Piping Pipe Justification Pipe Sizing Pipe add insulation &remove insulation Pipe Pressure loss report	Theory/Demo	4.0Hrs
30	Session 33,34	Exercises included for Practice	Practicals	4.0Hrs
		Fire Protection: Point of Connection		

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31	Session 35	Fire Pump Assembly Fire Riser Assembly Sprinkler Heads **Creating Fire Protection Systems:** Creating a Fire Protection Wet System Filtering Fire Protection Systems Fire Protection Pipe Settings and Routing	Theory/Demo	2.0Hrs
32	Session 36	Exercises included for Practice	Practicals	2.0Hrs
33	Session 37	**DRAFTING AND DETAIL TOOLS** Line Styles Regions Detail Components CAD Details Drafting Views Detail Library Detail Views	Theory/Demo	2.0Hrs
34	Session 38	Exercises included for Practice	Practicals	2.0Hrs
35	Session 39	**Sheets:** Creating a Title block Using Existing CAD Graphics Sheets in a Project File Placing Views on Sheets Viewports Properties Annotations Schedules Sheet Revisions Printing Sheets Export	Theory/Demo	2.0Hrs
36	Session 40	Exercises included for Practice	Practicals	2.0Hrs